

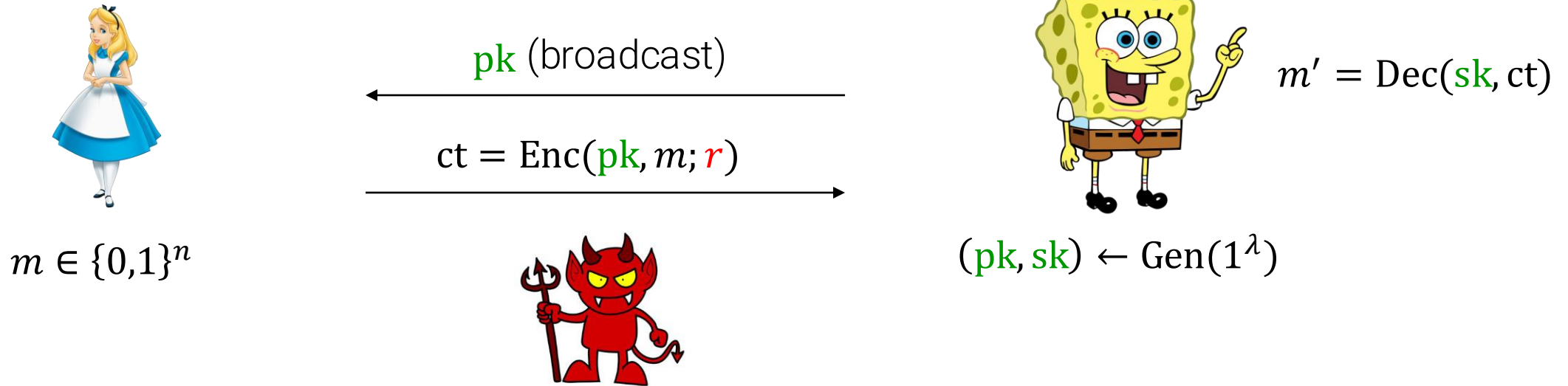
COS 433: Cryptography

Announcements:

- Problem Set 3 due [Wed 10/29](#)
- Problem Set 4 due [Wed 11/12](#)

Digital signatures (authentication)

Public-Key Encryption



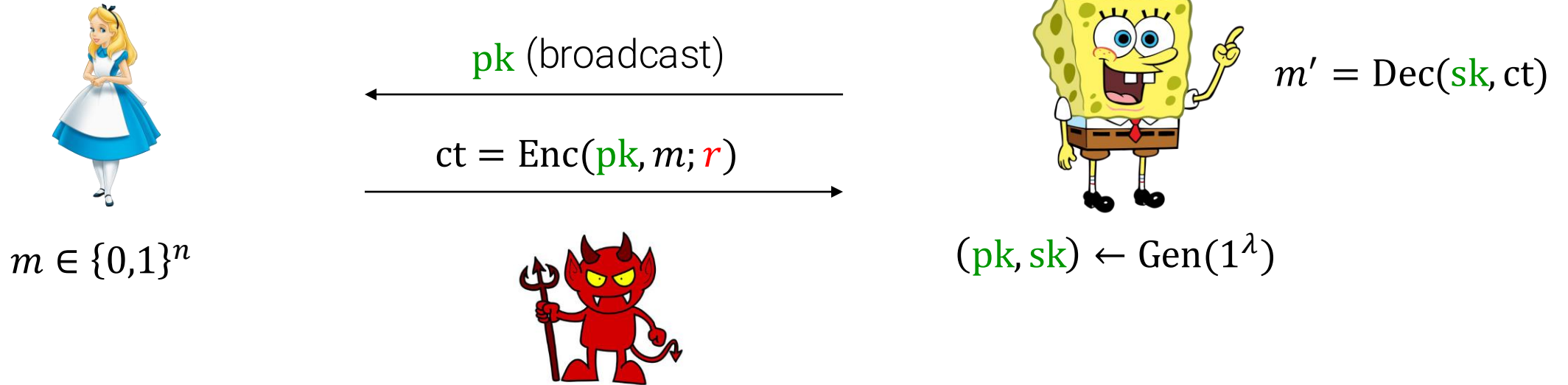
Problem: man-in-the-middle attacks

If pk broadcast message is tampered with, there is no hope.

Need: a public-key infrastructure. A process indicating that pk_{Bob} belongs to some entity named "Bob."

- We will assume that pk has been recorded to belong to Bob somehow.

Public-Key Encryption

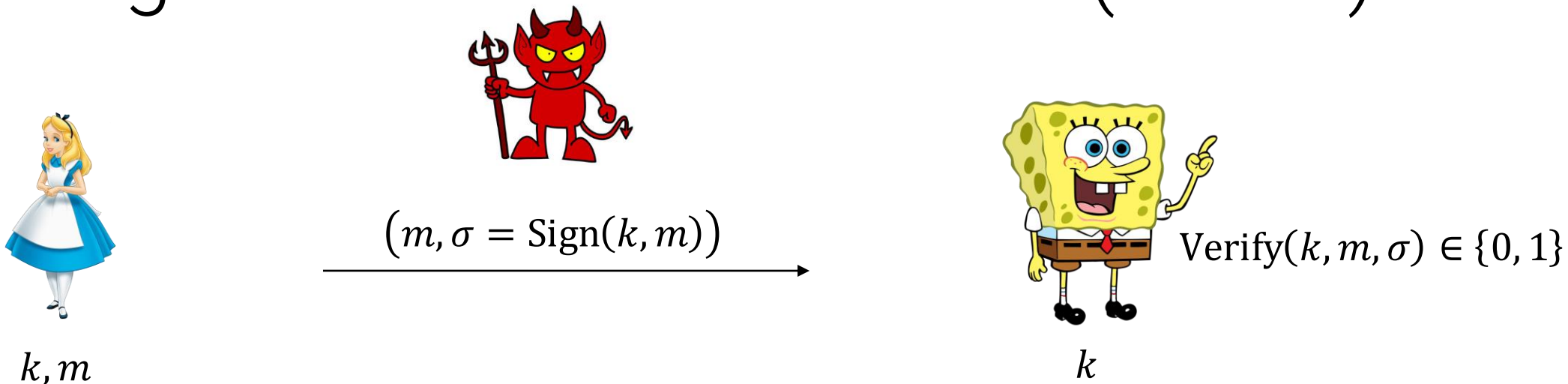


Problem: man-in-the-middle attacks

Main problem: how does Bob know a message was sent **by Alice**?

Bob may know a public key belonging to Alice, but so does the adversary!

Message Authentication Codes (MACs)



Definition: A **message authentication code (MAC)** scheme with message length $n = n(\lambda)$ consists of efficient algorithms **Sign**, **Verify**. It must satisfy the following properties.

- **Correctness:** for all k, m , $\text{Verify}(k, m, \text{Sign}(k, m)) = 1$ with probability 1.
- **Unforgeability:** for all efficient oracle adversaries $A^{\text{Sign}(\textcolor{red}{k}, \cdot)}$, we have:

$$\Pr[A^{\text{Sign}(\textcolor{red}{k}, \cdot)}(1^\lambda) \rightarrow (m^*, \sigma^*): m^* \text{ was not queried and } \text{Verify}(\textcolor{red}{k}, m^*, \sigma^*) = 1] = \text{negl}(\lambda)$$

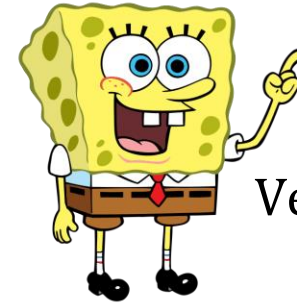
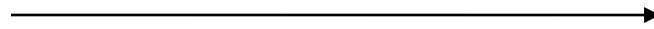
Digital Signatures



k, m



$(m, \sigma = \text{Sign}(k, m))$



k

$\text{Verify}(k, m, \sigma) \in \{0, 1\}$

We want: “public-key MACs”

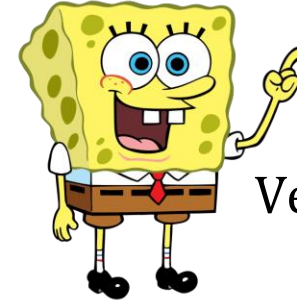
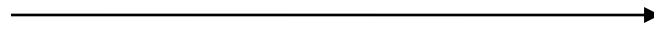
Digital Signatures



k, m



$(m, \sigma = \text{Sign}(k, m))$



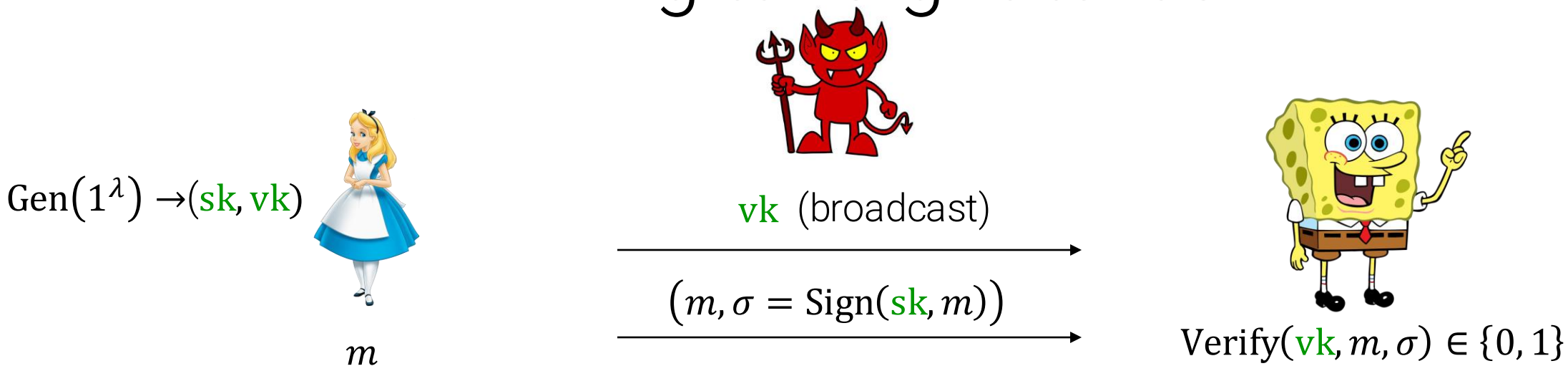
k

$\text{Verify}(k, m, \sigma) \in \{0, 1\}$

Whose key should be public?

Bob's (verification) key

Digital Signatures

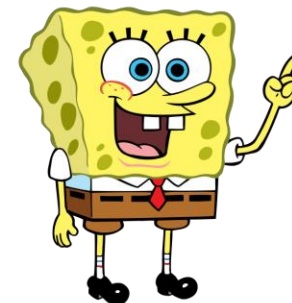


Definition: A **digital signature** scheme with message length $n = n(\lambda)$ consists of efficient algorithms Gen, Sign, Verify. It must satisfy the following properties:

- **Correctness:** for all m , $\text{Verify}(\text{vk}, m, \text{Sign}(\text{sk}, m)) = 1$ with probability 1.
- **Existential Unforgeability:** for all efficient oracle adversaries $A^{\text{Sign}(\text{sk}, \cdot)}$, we have:

$$\Pr_{(\text{sk}, \text{vk})} [A^{\text{Sign}(\text{sk}, \cdot)}(\text{vk}) \rightarrow (m^*, \sigma^*): m^* \text{ not queried and } \text{Verify}(\text{vk}, m^*, \sigma^*) = 1] = \text{negl}(\lambda)$$

Digital Signatures



$\text{Gen}(1^\lambda) \rightarrow (\text{sk}, \text{vk})$



$m \in \{0,1\}^n$

vk (broadcast)

$(m, \sigma = \text{Sign}(\text{sk}, m))$

$\text{Verify}(\text{vk}, m, \sigma) \in \{0, 1\}$

Notes:

- **Message length matters.** Not clear how to go from signing one-bit messages to signing longer messages.
- **Oracle-based security definition** is still required, since the signing key is secret.

$$\Pr_{(\text{sk}, \text{vk})} [A^{\text{Sign}(\text{sk}, \cdot)}(\text{vk}) \rightarrow (m^*, \sigma^*): m^* \text{ not queried and } \text{Verify}(\text{vk}, m^*, \sigma^*) = 1] = \text{negl}(\lambda)$$

“Textbook” RSA Signatures

$sk = (N, d)$
 $vk = (N, e)$



$m \in \mathbb{Z}_N^\times$



(N, e)

$(m, \sigma = m^d \bmod N)$



$\text{Verify}((N, e), m, \sigma) = 1$ if $\sigma^e \equiv m \pmod{N}$

Notes:

- Restricted to $m \in \mathbb{Z}_N^\times$
- **Does not satisfy** existential unforgeability. Why?
 - Can multiply any two signatures together to get a new one!
- **Secure** if the queries and challenge m^* are all **random** and **outside the adversary's control**.

“Textbook” RSA Signatures

$sk = (N, d)$
 $vk = (N, e)$

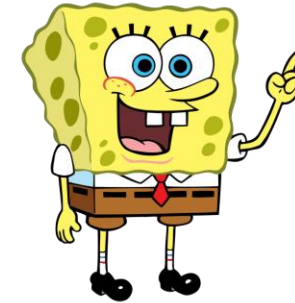


$m \in \mathbb{Z}_N^\times$



(N, e)

$(m, \sigma = m^d \bmod N)$



$\text{Verify}((N, e), m, \sigma) = 1$ if $\sigma^e \equiv m \pmod{N}$

Random Message Unforgeability: for all efficient adversaries $A^{\text{Sign}(sk, \cdot)}$ of the form



m^*, vk

“help me”

$m, \text{Sign}(sk, m)$



$$\Pr_{(sk, vk)} \left[A^{\text{Sign}(sk, \cdot)}(m^*, vk) \rightarrow \sigma^* : \text{Verify}(vk, m^*, \sigma^*) = 1 \right] = \text{negl}(\lambda)$$

“Textbook” RSA Signatures

$$\begin{aligned} sk &= (N, d) \\ vk &= (N, e) \end{aligned}$$

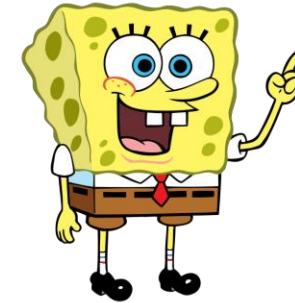


$$m \in \mathbb{Z}_N^\times$$



$$(N, e)$$

$$(m, \sigma = m^d \bmod N)$$



$$\text{Verify}((N, e), m, \sigma) = 1 \text{ if } \sigma^e \equiv m \pmod{N}$$

Proof of random message unforgeability: we want



$$m^*, N, e$$

“help me”

$$m, m^d$$



$$\Pr_{(sk, vk)} [A^{\text{Sign}(sk, \perp)}(m^*, N, e) \rightarrow (m^*)^d] = \text{negl}(\lambda)$$

“Textbook” RSA Signatures

$sk = (N, d)$
 $vk = (N, e)$



$m \in \mathbb{Z}_N^\times$



(N, e)

$(m, \sigma = m^d \bmod N)$



$\text{Verify}((N, e), m, \sigma) = 1$ if $\sigma^e \equiv m \pmod{N}$

Key Fact:

$$(m, m^d) \equiv (\sigma^e, \sigma)$$

"Textbook" RSA Signatures

$sk = (N, d)$
 $vk = (N, e)$

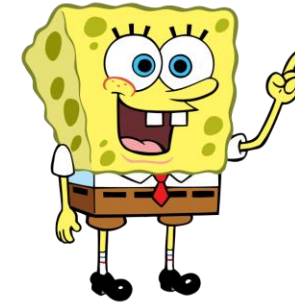


$m \in \mathbb{Z}_N^\times$



(N, e)

$(m, \sigma = m^d \bmod N)$



$\text{Verify}((N, e), m, \sigma) = 1$ if $\sigma^e \equiv m \pmod{N}$

Proof of random message unforgeability: equivalently, we want



m^*, N, e

"help me"

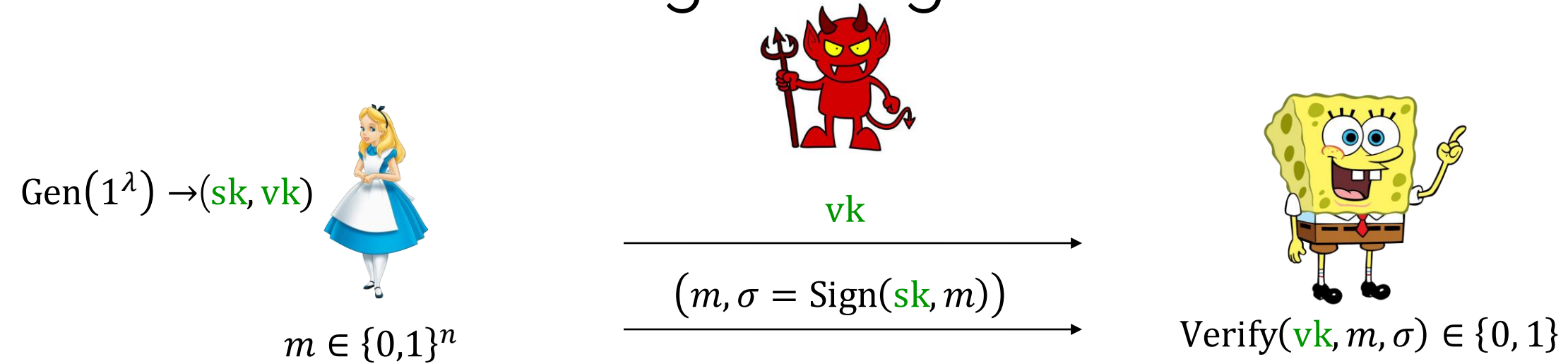
σ^e, σ

sk not needed here!

Breaks RSA assumption

$$\Pr_{(sk, vk)} [A^{\text{Sign}(sk, \perp)}(m^*, N, e) \rightarrow (m^*)^d] = \text{negl}(\lambda)$$

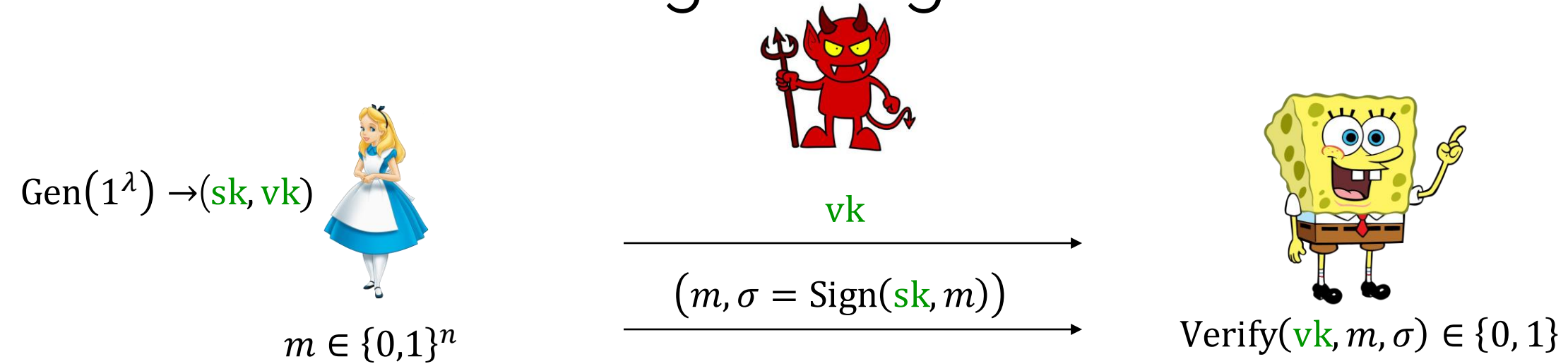
Digital Signatures



Building full-fledged digital signatures is pretty difficult!

- Practical constructions are usually **heuristically secure**
- But we do know how to provably build signature schemes (under assumptions)

Digital Signatures

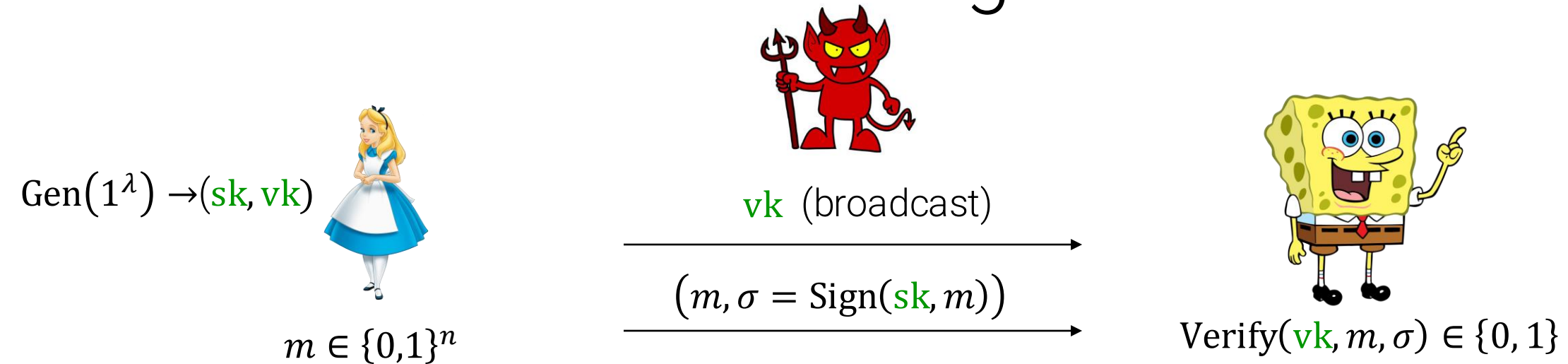


Today and next two classes:

1. One-time signatures (easier to build)
2. RSA signatures (heuristic)
3. DSA/Schnorr signatures (heuristic)

One-time signatures

One-Time Signatures

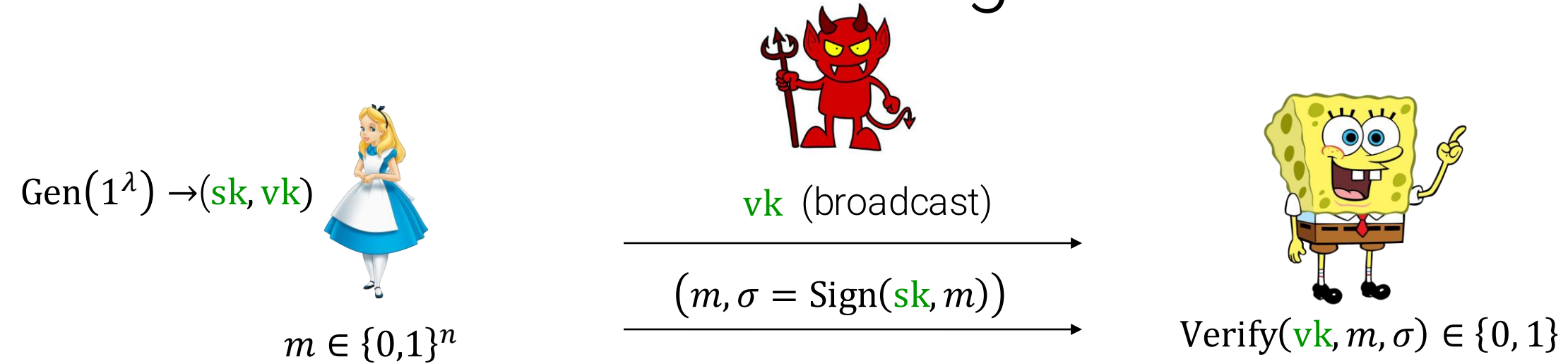


Definition: A **one-time digital signature** scheme with message length $n = n(\lambda)$ consists of efficient algorithms $\text{Gen}, \text{Sign}, \text{Verify}$. It must satisfy **correctness** as well as:

- **One-time Existential Unforgeability:** for all efficient oracle adversaries $A^{\text{Sign}(\text{sk}, \cdot)}$ that **query the signing oracle once**, we have:

$$\Pr_{(\text{sk}, \text{vk})} \left[A^{\text{Sign}(\text{sk}, \cdot)}(\text{vk}) \rightarrow (m^*, \sigma^*) : m^* \text{ not queried and } \text{Verify}(\text{vk}, m^*, \sigma^*) = 1 \right] = \text{negl}(\lambda)$$

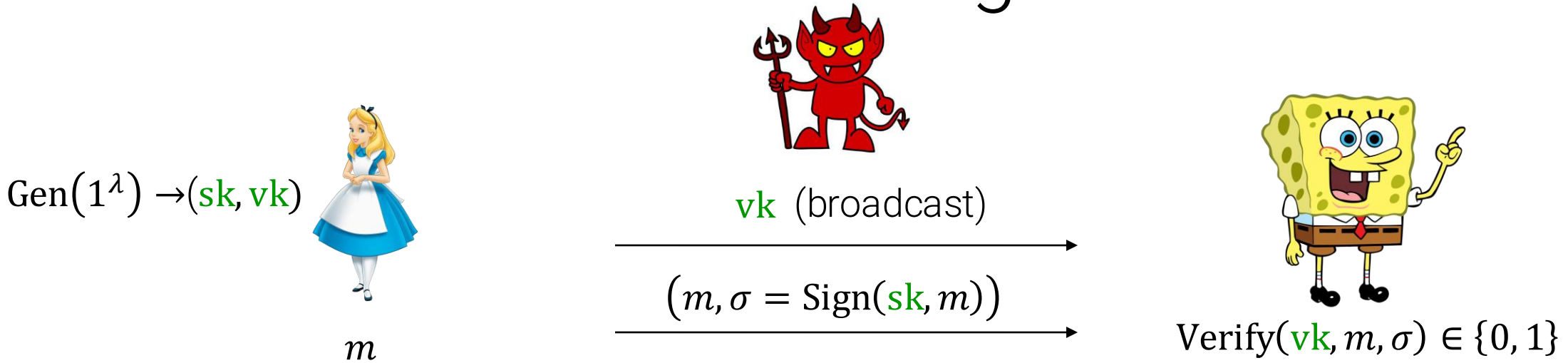
One-Time Signatures



Theorem: Assuming one-way functions exist, for any $n(\lambda) = \text{poly}(\lambda)$, there exist one-time signatures with message length $n(\lambda)$!

- **Key length:** $|vk| = n(\lambda) \cdot \lambda > n(\lambda)$ (long keys)
- We will fix this later
- We don't need any number theory!

“Zero-Time Signatures”

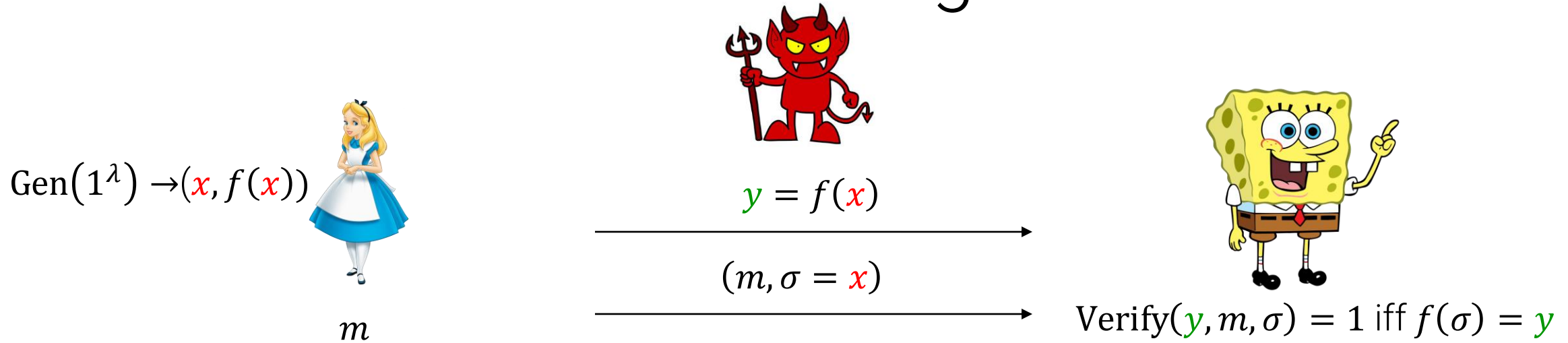


Definition: A “zero-time” digital signature scheme with message length $n = n(\lambda)$ consists of efficient algorithms Gen , Sign , Verify . It must satisfy **correctness** as well as:

- “Zero-time” **Existential Unforgeability**: for all efficient adversaries $A(\text{vk})$, we have:

$$\Pr_{(\text{sk}, \text{vk})} [A(\text{vk}) \rightarrow (m^*, \sigma^*): \text{Verify}(\text{vk}, m^*, \sigma^*) = 1] = \text{negl}(\lambda)$$

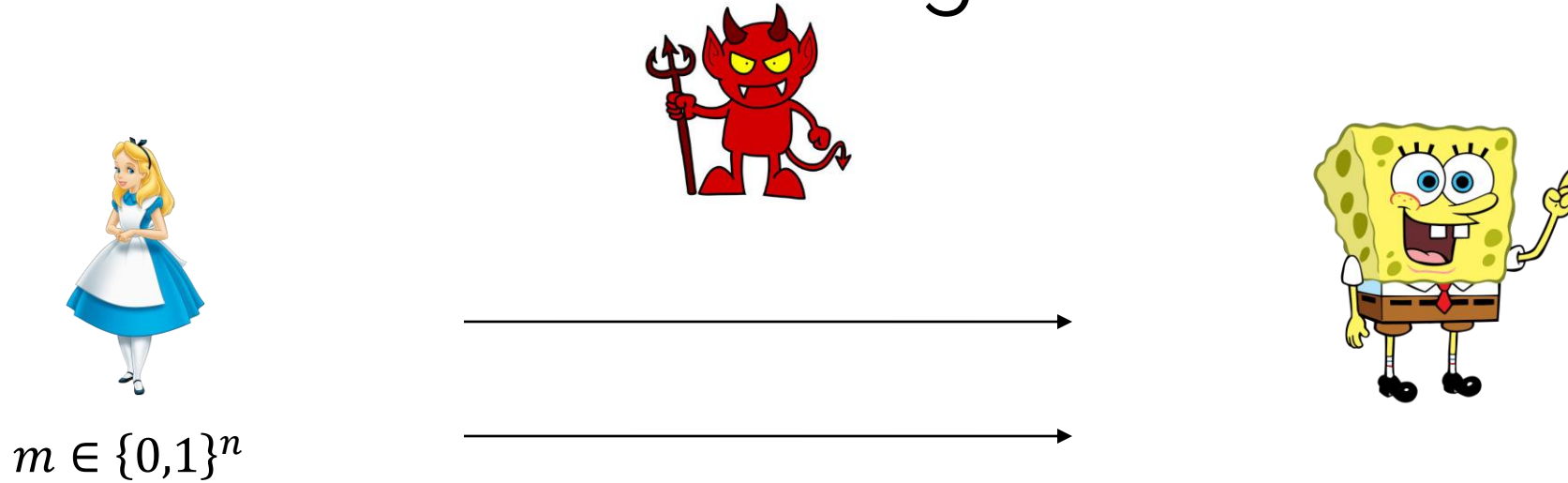
“Zero-Time Signatures”



Construction from any OWF:

- $\text{Gen}(1^\lambda) \rightarrow (\mathbf{x}, \mathbf{y} = f(\mathbf{x}))$
- To sign a message, send $\mathbf{x}$. To verify, check that σ is a valid preimage of $\mathbf{y}$.
- **Unforgeability** follows immediately from one-wayness of f

One-Time Signatures



Each m now needs to have its **own distinct signature**.

One-Time Signatures

$$\text{Gen}(1^\lambda) \rightarrow \left((x_{i,b}), (f(x_{i,b})) \right)$$

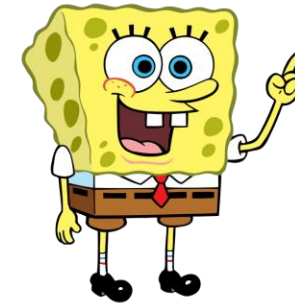


$$m \in \{0,1\}^n$$



$$(y_{i,b})_{i,b}$$

$$(m, \sigma = (x_{i,m_i})_i)$$



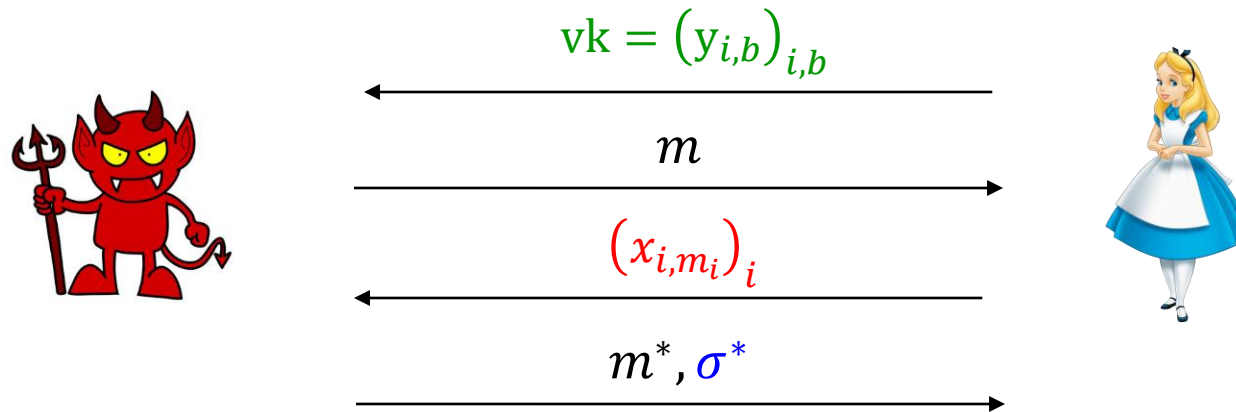
$$\text{Verify}(\mathbf{vk}, m, \sigma) = 1 \text{ iff } f(\sigma_i) = y_{i,m_i}$$

Construction from any OWF:

- $\text{Gen}(1^\lambda) \rightarrow \left((x_{i,b})_{i \in [n], b \in \{0,1\}}, (f(x_{i,b}))_{i \in [n], b \in \{0,1\}} \right)$ ($2n$ input/output pairs)
- To sign a message, send n preimages: x_{i,m_i} for all i .
- To verify, check the n preimages are all valid.

One-Time Signatures

Security Proof: suppose that



$$\Pr_{(sk, vk)} [A^{\text{Sign}(sk, \cdot)}(vk) \rightarrow \text{valid } m^* \neq m, \sigma^*] = \epsilon$$

e.g.: $m = (0, 0, 1, 0, 0, 1, 1, 1)$

$m_3^* = 0$

One-Time Signatures

Security Proof:

$$y = f(x)$$



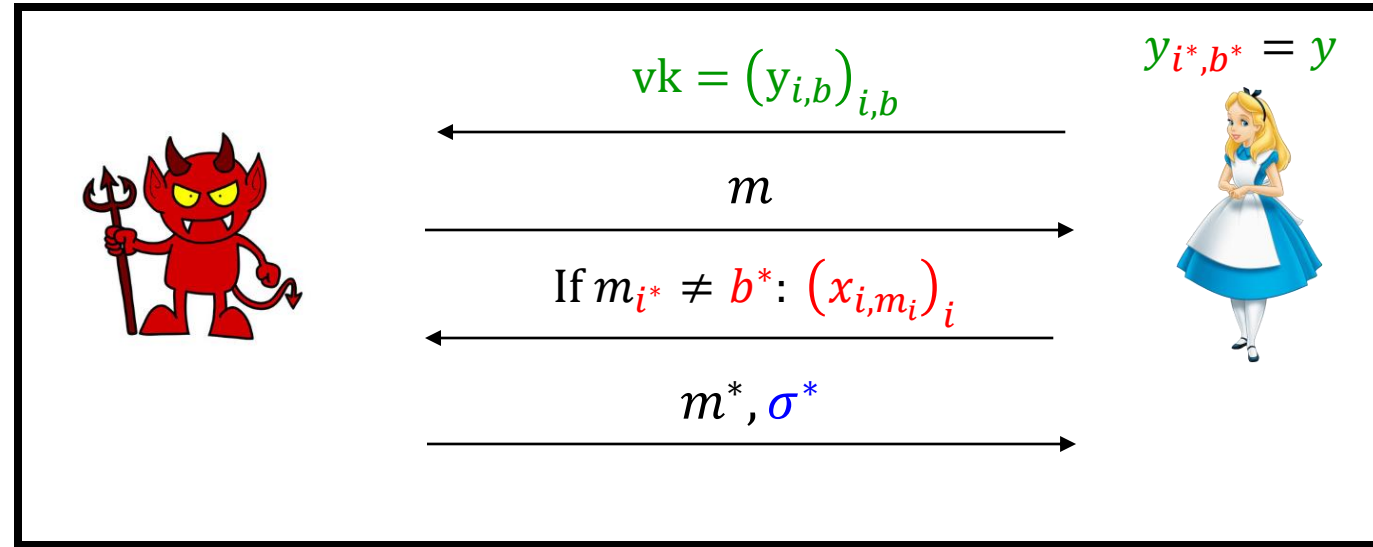
sample i^*, b^*

$\sigma_{i^*}^*$

Get m, m^*, σ^*

$$\Pr[f(\sigma_{i^*}^*) = y] \geq \frac{\epsilon}{2n}$$

- Emulation succeeds with probability 1/2
- Conditioned on forgery, $m_{i^*}^* = b^*$ with probability $\geq \frac{1}{n}$ (since $m^* \neq m$)



continued next time